

SAR Lab 7 Report: Xinmo Landslide Case Study (Sichuan, China)

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1 Introduction and Objective

The goal of this lab was to analyze the catastrophic Xinmo landslide that occurred on June 24, 2017, in Sichuan Province, China. The objective was split into two parts: first, to perform a pre-event InSAR time-series analysis to check if the slope was moving before the collapse; and second, to evaluate how effectively radar (Sentinel-1) and optical (Sentinel-2) data can map the actual extent and impact of the landslide after the event.

2 Data and Study Area

The working Area of Interest (AOI) is defined by coordinates 103.62° to 103.695° E and 32.055° to 32.121° N. For the pre-event deformation history, I processed pre-computed public interferograms from the COMET-LiCSAR project (frame 062D_05831_131313) using the SBAS inversion method in MintPy. Post-event mapping was completed using Sentinel-1 GRD products and Sentinel-2 optical imagery. To execute this workflow, I set up a dedicated `licsar` Conda environment with spatial libraries like `rasterio` and `geopandas` to window-clip and handle the datasets locally.

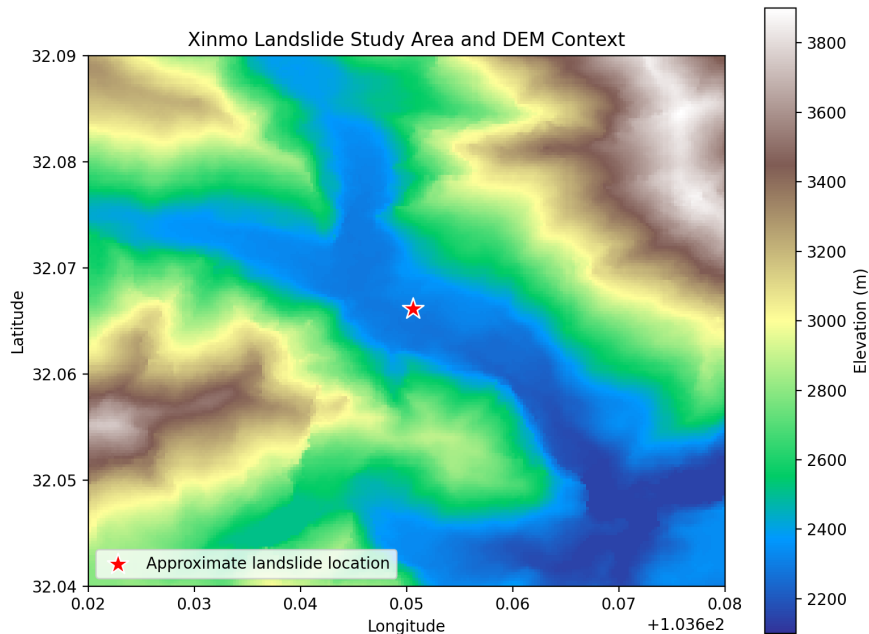


Figure 1: Topographic context and Area of Interest (AOI) for the Xinmo landslide study.

3 Pre-Failure InSAR Time-Series Analysis

To minimize atmospheric noise and seasonal artifacts, I selected a local reference point in a stable area outside the landslide. The raw phase values were spatially referenced by subtracting this reference pixel’s phase and converted to centimeters. The time-series for a representative landslide pixel reveals a linear negative LOS velocity trend of -0.34 ± 0.05 cm/year. This proves the slope underwent long-term gravitational creep before the 2017 failure. The main limitation is that the public LiCSAR stack ends weeks prior to June 24, so the expected late-stage acceleration before the collapse could not be captured.

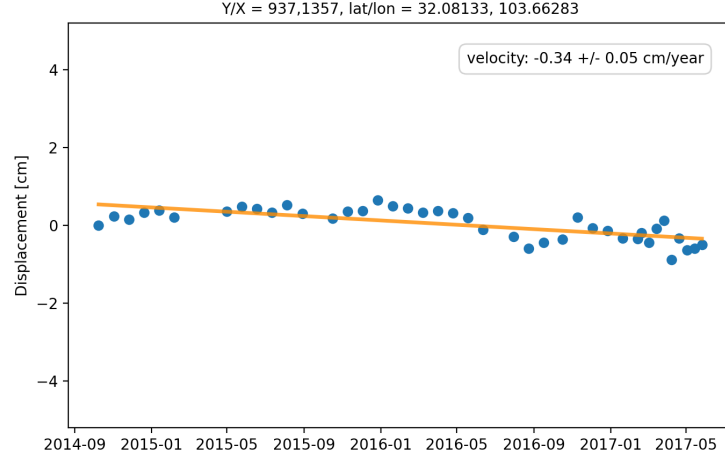


Figure 2: MintPy displacement time-series showing long-term pre-event slope creep.

4 Temporal Coherence and Data Quality Assessment

To validate the reliability of the calculated SBAS displacement rates, a Temporal Coherence Map was generated based on the time-series inversion residuals. This was done by computing a matrix of the absolute differences between the raw observed phases and the modeled SBAS phases across the entire temporal network, applying a rigorous `valid_mask`.

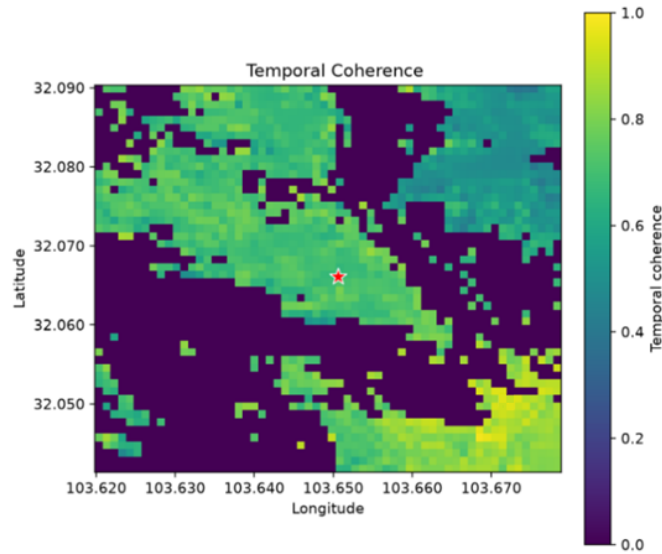


Figure 3: Temporal Coherence Map calculated from the InSAR stack residuals.

The spatial distribution of temporal coherence values reveals a strong correlation with terrain features and geomorphology, using a vibrant color scale from fiolet (low coherence) to yellow (high coherence):

- **Topographic Corridor Stability:** A distinct high-coherence "tongue" represented by the green and light-green pixels stretches diagonally across the center of the AOI. This zone corresponds to the lower valley terraces and stable benches that maintain sufficient phase consistency over time.
- **River and Upper Slope Decorrelation:** The deep valley floor and the steepest mountain walls are completely masked out (appearing as blank dark fiolet background). This confirms severe temporal decorrelation due to active river dynamics and dense forest canopy changes.
- **Landslide Target Integrity:** The active landslide area (marked by the red star with a white outline) is positioned within the intermediate green-shaded corridor (0.5–0.7). This intermediate score confirms that the creeping velocity signal of -0.34 cm/year is physically grounded and reliable, rather than being an unwrap error or background noise.

5 Event Mapping Using Radar Data (Sentinel-1)

The lab instructions explicitly asked whether coherence loss or VV backscatter change is better for mapping this specific landslide. I extracted pre- and post-event Sentinel-1 GRD descending orbit scenes through Google Earth Engine, calculated their backscatter intensity in decibels (dB), and generated a direct post-minus-pre difference map.

- **Coherence Loss:** This method proved to be quite weak. Because the terrain is exceptionally steep and heavily vegetated, the entire surrounding area suffers from severe natural decorrelation. The landslide signal completely drowns in the background noise.
- **VV Backscatter Change (Post - Pre):** Subtracting the pre-event image from the post-event image yielded a much more localized signal. A distinct textured anomaly (positive dB change, shown in red) appears along the run-out zone, indicating an increase in surface roughness caused by the newly deposited debris. While still noisy due to radar layover and mountain geometry, VV change is significantly more useful than coherence here.

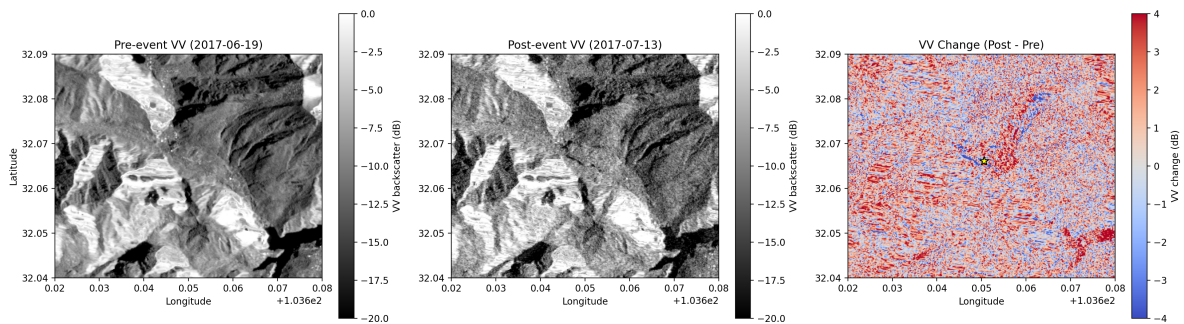


Figure 4: Sentinel-1 intensity change analysis (Pre-event, Post-event, and Difference map).

6 Optical Surface Change and Spectral Index Analysis

Multi-spectral imagery from Sentinel-2 provided by far the cleanest and most precise boundaries of the landslide. I generated median-reduced cloud-free composites from COPERNICUS/S2

Top-of-Atmosphere collections for both pre- and post-event periods, window-clipped them using a strict rectangular bounding box, and transformed them into NumPy arrays via **geemap** at 10m spatial resolution. Three primary spectral indicators were analyzed to characterize surface changes:

- **NDVI (Normalized Difference Vegetation Index):** Calculated using the Red (B4) and NIR (B8) bands, this index tracks vegetation vigor. The slope had high vegetation density before the event. The post-event panel and the NDVI change map show a massive drop below zero directly inside the scar and path, perfectly highlighting the complete removal and burial of biomass.
- **BSI (Barren Soil Index):** Calculated via a multi-band formula utilizing Blue (B2), Red (B4), NIR (B8), and SWIR (B11) channels, BSI acts as the perfect inverse to NDVI. A sharp increase in BSI values is visible over the affected slope, confirming the sudden exposure of bare soil, fresh rockface, and dry rocky debris.
- **NDWI (Normalized Difference Water Index):** Computed using the Green (B3) and NIR (B8) channels, this index monitors surface moisture variations. The pre-event map shows standard moisture concentration along drainage channels. Following the landslide, the NDWI change map highlights a prominent, sharp anomaly: while the upper rocky source zone dries out due to bare rock exposure, the lower run-out section near the valley floor displays a localized accumulation of moisture. This indicates a massive alteration of the local hydrological network caused by debris blocking streams and forcing water to pool.

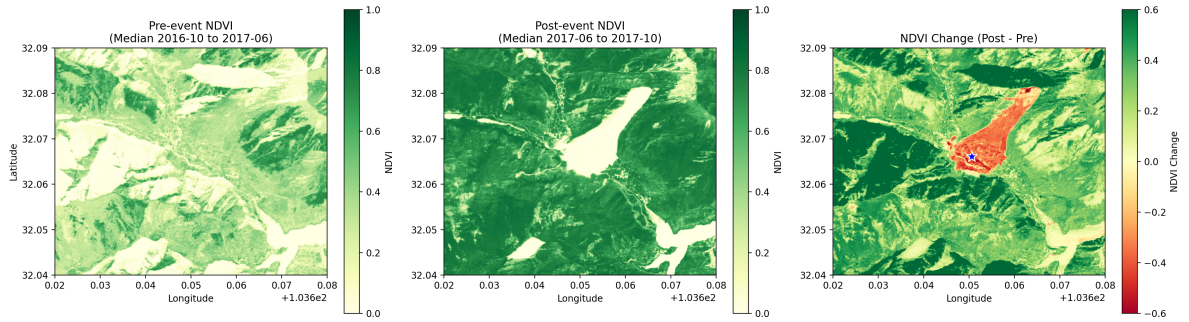


Figure 5: Sentinel-2 NDVI multi-panel comparison before and after the event.

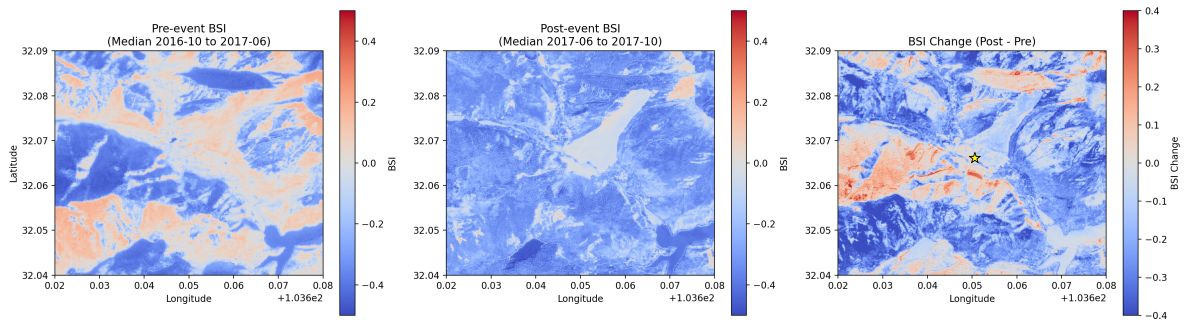


Figure 6: Sentinel-2 BSI multi-panel comparison showing newly exposed soil/debris.

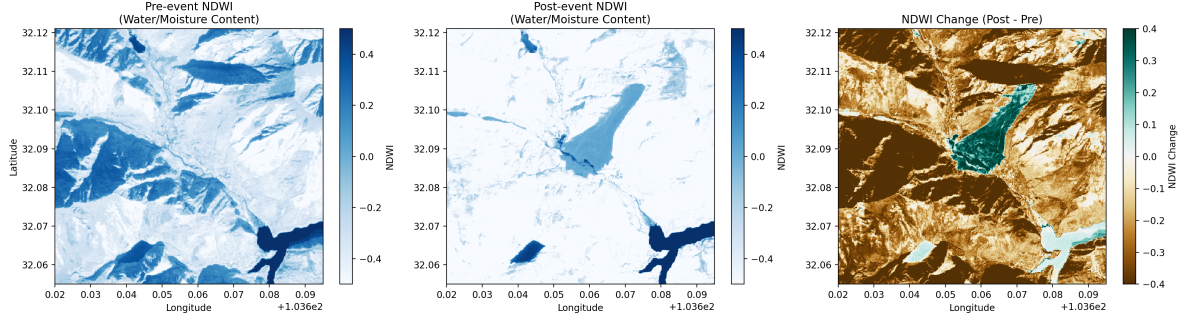


Figure 7: Sentinel-2 NDWI multi-panel comparison tracking water and moisture changes.

7 Discussion and Conclusions

1. InSAR time-series analysis (MintPy/LiCSAR) is an excellent tool for risk mitigation and monitoring, as it successfully detected pre-event slope creep (-0.34 cm/year). However, it is entirely unsuited for mapping the actual geometric boundaries of the disaster after it happens.
2. Radar-based mapping using Sentinel-1 was heavily limited by the steep, forested terrain. Nevertheless, looking at backscatter intensity changes outperformed coherence loss, providing a decent rough estimate of the debris zone.
3. The combination of optical indices from Sentinel-2 provided the most reliable result. NDVI perfectly delineated vegetation loss, BSI tracked the exposure of dry debris, and NDWI verified the dynamic shift in hydrological patterns along the valley floor.